INCH-POUND
A-A-60028
June 6, 2022

COMMERCIAL ITEM DESCRIPTION

COUNTER ELECTRONIC FREQUENCY

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This electronic counter is a high performance counter for measurement, analysis and calibration of frequency, time interval and phase.

2. CLASSIFICATION. The electronic counter covered by this commercial item description (CID) shall be commercially available equipment that has factory-installed options and/or may be modified to the extent necessary to meet the following description. The electronic counter shall be class 3, in accordance with MIL-PRF-28800.

2.1 Order of precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. SALIENT CHARACTERISTICS.

3.1 Item description. Electronic counter is a frequency, period, pulse, “dead time” and event counter that can measure discrete frequencies, phase, duty cycle, rise time, event analytics with a histogram and time periods or intervals in a gated or continuous form.

3.2 Reference clock. The electronic counter shall have a Rubidium atomic reference clock.

3.3 Frequency. The electronic counter shall have a minimum operating frequency range from 0.002 hertz (Hz) to 3.0 gigahertz (GHz).

3.4 Resolution. The electronic counter shall have a minimum resolution of 12 digits per second measuring time (normal mode resolution).

3.5 Sample speed. The electronic counter shall have a minimum sample speed of 250 kilo sample per second (ks/s).

3.6 Memory depth. The electronic counter shall have a minimum memory depth of 3.75 million readings.

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3.7 Frequency burst. The electronic counter shall be able to read a burst (gated/window) duration with a minimum duration of 40 ns.

3.3 Repetition frequency. The electronic counter shall have a pulse repetition frequency (PRF) sampling range from 0.5 hertz (Hz) to 1.0 megahertz (MHz).

3.9 Time resolution. The electronic counter shall have a time resolution for measurement of a single burst with a duration ≥ 50 pico seconds (ps).

3.10 Internal time-base aging. The internal-reference frequency aging variants shall be

$\leq \pm 2x10^{-10}$ per year and power-on stability deviation after 24 hours on time $\leq 5x10^{-10}$.

3.11 Operating input voltage low frequency range. The electronic counter shall have an input voltage range from 10 mV_{RMS} to 2.5 V_{RMS}, frequency range from 0 to 400 megahertz (MHz) for input A, B.

3.12 Operating input voltage high frequency range. The electronic counter shall have an input voltage range from 10 mV_{RMS} to 2.5 V_{RMS}, frequency range from 500 MHz to 3.0 gigahertz (GHz) for input C @ 50 ohm (Ω).

3.13 Ratio. The electronic counter shall have analysis functions with a ratio range from 10^{-9} to 10^{11}.

3.14 Multi-parameter. The electronic counter shall have auxiliary measurements values such as $V_{max}/V_{min}/V_{P-P}$ in frequency, attenuation and phase.

3.15 Display. The electronic counter shall have an alpha-numerical display showing the frequency, period, time, pulse, phase, range, input label tags, analytic & parameters and error codes when present. It shall have 14 digits in numerical mode and be a high contrast unit.

3.16 Design and construction. The electronic counter shall be used in both bench-top and rack-mount environments. The electronic counter is a commercial item subject to commercial-product regulatory standards. The electronic counter shall be designed and constructed in accordance with all applicable European Union (EU) requirements with the "CE" marking affixed, International Organization for Standardization (ISO), the International Electro-technical Commission (IEC), the International Telecommunication Union (ITU), Underwriters Laboratories (UL), and the American National Standards Institute (ANSI). The electronic counter shall be compliant with the Restriction of Use of Hazardous Substances (RoHS) and Waste from Electrical and Electronic Equipment (WEEE) guidelines and policies. It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.17 System safety. The electronic counter shall comply with all sections of the UL 1449 standard for electrical safety and the IEC 61010-1.

3.18 Sound levels. The electronic counter acoustic noise shall conform to sound-level requirements of ISO 7779 and measured in accordance with ISO 8253, section 2.

3.19 Electrostatic discharge (ESD). The design of the electronic counter shall preclude equipment damage due to electrostatic discharge (ESD) by incorporating mitigating methods such as shielding, grounding and the use of materials that minimizes electrostatic charge buildup and sparking in accordance with ANSI/ESD-S20.20-2014, section 7.0, ESD Control Program.

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3.20 Electromagnetic interference (EMI). The electronic counter's EMI emissions shall be in accordance with the radiated-emission and susceptibility requirements of MIL-PRF-28800; MIL-STD-461, sections RE102, RS103; IEC 1000-2-5; and the emitted radiation absorption and radiation limits of CFR, Title 47, Parts 15.109 and 15.209.

3.21 Service life. The design of the electronic counter shall be such that, under normal use and operation, the service repair can be performed according to MIL-PRF-28800, sections 3.15.2 and 3.15.3.

3.22 Port damage. The inputs ports shall withstand up to 12 volts root-mean-square (V_{RMS}) for inputs A and B without damaging the internal circuitry. The input port "C" shall withstand up to 2.5 V_{RMS} without damaging the internal circuitry.

3.23 Connectors. The system shall have bayonet Neill-Concelman (BNC)-type connectors for other front- and rear-panel inputs and outputs. Type "N" female connector. Subminiature version A (SMA) female input. A general-purpose-interface-bus (GPIB) interface with an IEC 60488 (formerly IEEE-488.2) protocol connector is required for compatibility and control functions.

3.24 Operating-temperature range. The electronic counter shall meet the specifications listed herein with full capabilities within the temperature range of 32 to 113 °F (0 to 45 °C).

3.25 Storage-temperature range. The electronic counter shall meet or exceed manufacturer's specifications for stored equipment within the temperature range of -40 to 160°F (-40 to 71°C) after tests of MIL-PRF-28800, section 4.5.5.1, have been performed for instruments in the range of class 3 equipment classification per MIL-PRF-28800, table XI.

3.26 Operating-humidity range. The electronic counter shall meet its performance and accuracy requirements for humidity conditions as specified in MIL-PRF-28800, section 3.9.2.3 within the humidity range of 5 percent-95 percent 50 to 86 °F (10 to 30 °C).

3.27 Altitude requirements. The electronic counter shall meet its operating performance and accuracy while operating at an altitude of 15,000 feet (4,600 meters). The equipment shall meet its performance and accuracy requirements for operating and non-operating storage as specified in MIL-PRF-28800, sections 3.9.3 and 4.5.5.2 respectively, at this altitude.

3.28 Heat dissipation and ventilation. Ventilation shall be sufficient to allow for proper heat dissipation that can be achieved by either passive or active methods in order to meet MIL-PRF-28800, section 4.5.5.1, under temperature and humidity tests for class 3 equipment classification per MIL-PRF-28800, table XI.

3.29 Calibration and maintenance adjustments. No special tools, software, technical manuals, or documentation shall be required for the alignment, adjustment, repair, and calibration of the electronic counter, except those provided with the electronic counter. A software license shall be issued for each physical instrument when the system is designed for computer-based calibration. Software licenses needed to perform the alignment/adjustment process (where required by the manufacturer) shall be valid for the life of the instrument and shall be included with the electronic counter. Any special or unique accessories not common to a secondary-calibration laboratory required for calibration of this item shall be furnished by the manufacturer, to include, but not limited to, software, fixtures, tools, and standards.

3.30 Calibration period. The design of the electronic counter shall provide readily accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without

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removal of the test-set case or modules. The calibration interval shall be based on a period of one calendar year or an operating interval of 2,080 hours, whichever occurs first. At the end of this interval, 85 percent of the units shall remain within calibration tolerances in accordance with MIL-PRF-28800, section 3.13.

3.31 Reliability. The design of the electronic counter shall be such that under normal use and operation, the mean time between failures (MTBF) shall exceed 2,080 hours of operation with a statistical certainty equal to or greater than 95 percent in accordance with MIL-PRF-28800, section 3.14, based on an 8-hour workday, 40-hour week, and 52-week interval. The manufacturer shall be able to reasonably establish the claim and certify with NIST traceability that the offered equipment is capable of the required level of reliability by meeting or exceeding MIL-PRF-28800, sections 3.14 and 4.5.8.

3.32 Built-in test equipment and diagnostics. The electronic counter shall be equipped with built-in test and diagnostic capability that performs a system test evaluating all functional characteristics of the instrument at “Power On” or “Cold Start” in accordance with MIL-PRF-28800, section 6.6.6.

3.33 Maintainability. The electronic counter shall meet the human-engineering factors of ergonomics for operational and maintainability criteria providing access to controls, internal components, and adjustments in accordance with MIL-STD-1472, sections 5.9 through 5.9.16, and MIL-PRF-28800, section 6.6.15.

3.34 Preventive maintenance. Preventive maintenance shall not require more than 15 minutes per 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed in accordance with MIL-PRF-28800, section 3.15.2.

3.35 Corrective maintenance. The electronic counter shall be designed to meet reparability criteria in accordance with MIL-PRF-28800, section 3.15.1.

3.36 Electrical power sources and connections. The electronic counter equipment shall operate from commercial, military, and shipboard power sources with a nominal voltage of 120 voltage root-mean-square (V_{RMS}), single-phase alternating current (AC) (±10 percent) at line frequencies of 50 Hz and 60 Hz, in accordance with MIL-PRF-28800, sections 3.6.1.2 and 3.6.1.2a.

3.37 Electrical surge. The electronic counter shall be able to provide surge protection that meets UL 1449, section 38; IEC 61000-4-4; and IEC 61010-1.

3.38 Noise suppression. The electronic counter shall be able to operate with electrical noises and harmonic content present on the alternating-current power source with frequencies in the 30 Hz to 50 kHz range and levels varying from line level of 41.6 dB ±1 dB (108 to 132 VAC) measured at the power input in accordance with UL 1283, IEC/EN 60939-1 and under the guidance of the IEEE-519 standard.

3.39 Dimensions. The electronic counters shall comply with the width and depth of the EIA/ECA-310-E standard equipment rack configuration.

3.40 Rack mount. The electronic counter shall be capable of being rack mounted in a standard EIA/ECA-310-E equipment rack and, if required, include a rack-mount conversion kit that meets the requirements of MIL-PRF-28800, section 3.7.4.13.3.

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3.41 Weight. The total weight of the electronic counter, excluding accessories and manuals, shall comply with MIL-STD-1472 for weight and size restrictions for single-person lift and handling, and shall not exceed a one-person lift ≤ 35 pounds (15.6 kilograms).

3.42 Markings. All warning decals and instrument-category ratings must conform to commercial safety standards and be visible by the user in accordance with commercial product-safety labeling regulations and MIL-STD-130. Markings shall be applied with paint, decals, engraving, or molding, and shall be the largest size possible. Warning/Danger text shall be lusterless red.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials or bio-based content to the maximum extent practicable, in accordance with Federal Acquisition Regulation, Part 23.403.

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. PRODUCT CONFORMANCE. Verification of the delivered electronic counter will be performed by the end user at the delivery point.

6. PACKAGING.

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Government documents. Copies of specifications, standards, and handbooks are available online at https://quicksearch.dla.mil/.

7.1.2 Code of Federal Regulations (CFR). Copies of CFRs may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402, or online at http://www.ecfr.gov/cgi-bin/ECFR?page=browse.

7.1.3 EOS/ESD Association. Copies of EOS/ESD standards may be obtained from the EOS/ESD Association, Inc. 7900 Turin Road, Building 3, Rome, NY 13440, or online at https://www.esda.org.

7.1.4 IEC Standards. Copies of ISO/IEC standards may be obtained from the IEC Regional Centre for North America (IEC-ReCNA) 446 Main Street, 16th Floor Worcester, MA 01608, or online at http://www.iec.ch/.

7.1.5 Institute of Electrical and Electronics Engineers (IEEE). Copies of these documents may be obtained from IEEE Operations Center, 445 Hoes Lane, Piscataway, NJ, 08854, or online at http://www.ieee.org.

7.1.6 Underwriters Laboratories (UL). Copies of these documents may be obtained from Underwriters Laboratories, 333 Pfingsten Road, Northbrook, IL 60062, or online at http://www.ul.com/.

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7.1.5 Federal Acquisition Regulation (FAR). Copies of FARS may be obtained at https://www.acquisition.gov/browse/index/far

7.2 Ordering data. Procurement documents must specify: title, number, date of this CID, the issue of the ASSIST Online (https://assist.dla.mil) to be used in the solicitation, and preservation and packaging requirements.

7.3 Key words.
Calibration.
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MILITARY INTERESTS

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